Laplace Transform

Mostly not absolutely integrable.

$u(t) \rightarrow$ approximation $(\delta(\omega))$

not a proper function

---

* $x(t) = e^{at} u(t)$ $\boxed{Re\{a\} < 0}$ ← for $X(\omega)$

$$x(t) \longrightarrow \boxed{h(t)} \longrightarrow y(t) = x(t) * h(t)$$

LTI

$$Y(\omega) = \underbrace{X(\omega)}_{\text{Not Defined}} H(\omega)$$

$\boxed{\text{\# Damping factor / Convergence factor}}$

(additional degree of freedom)

$x(t) \longrightarrow x(t) e^{-\sigma t}$

(may become absolutely integrable)

↳ (region of convergance)

$$\underbrace{e^{-\sigma t}}_{\text{convergence / damping factor}}$$

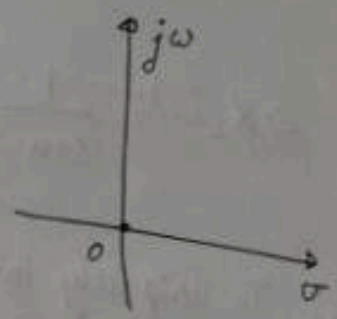
$$\Rightarrow \underline{L.T.} \{x(t)\} \triangleq F.T. \{x(t) e^{-\sigma t}\}$$

$$= \int_{-\infty}^{\infty} x(t) e^{-\sigma t} e^{-j\omega t} dt = \int_{-\infty}^{\infty} x(t) e^{-(\sigma + j\omega)t} dt$$

$$= \int_{-\infty}^{\infty} x(t) e^{-st} dt \triangleq X(s) \qquad \boxed{s \triangleq \sigma + j\omega}$$

$X(s) \triangleq \int_{-\infty}^{\infty} x(t) e^{-st} dt \Rightarrow$ } L.T. $\{x(t)\}$

$\Rightarrow \underset{1\text{-}D}{x(t)} \overset{L.T.}{\longleftrightarrow} \underset{2\text{-}D}{X(s)}$



$\Rightarrow$ F.T. $\{x(t)\}$ in terms of L.T. $\{x(t)\}$?

$\sigma = 0 \Rightarrow s = j\omega \Rightarrow X(s) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt = X(\omega)$

$$X(\omega) = X(s)\Big|_{\sigma=0}$$

$\Rightarrow$ Fourier Transformable

Ex: $x(t) = e^{2t} u(t)$

$$X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt = \underbrace{\int_{0}^{\infty} e^{2t} e^{-st} dt}_{} = \frac{1}{s-2}, \quad Re\{s\} > 2$$

$e^{(2-s)\infty} \to 0$

when $2 - Re\{s\} < 0$

$\Rightarrow Re\{s\} > 2$

(*) $X(s)$ has 2 parts:

i) Algebraic Expression

ii) ROC

Ex: $x_1(t) = e^{-at} u(t)$, $x_2(t) = -e^{-at} u(-t)$

$\Rightarrow X_1(s) = \int_0^{\infty} e^{-at} e^{-st} dt \Rightarrow e^{-(a+s)t}\Big|_{\infty} \to 0$ when

$= \frac{1}{s+a}$, $Re\{s\} > -a$

$$\Rightarrow X_2(s) = -\int_{-\infty}^{0} e^{-at} e^{-st}\,dt \Rightarrow e^{-(a+s)\infty} \to 0$$

when $Re\{s\} + a < 0$

$$\Rightarrow X_2(s) = \frac{1}{s+a}, \quad Re\{s\} < -a \qquad \Rightarrow Re\{s\} < -a$$

$\therefore$ To uniquely invert the L.T. $X(s)$ corresponding ROC must be specified.

* ROC:

(I) $s_0 \in ROC \Rightarrow X(s)\big|_{s_0} < \infty \Rightarrow F.T.\left\{x(t)\, e^{-\sigma_0 t}\right\} < \infty$

$= \sigma_0 + j\omega_0$

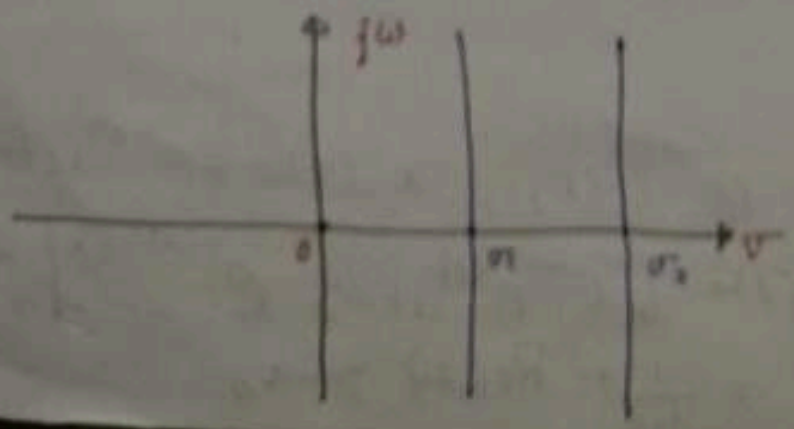
{ if FT exists $x(t)e^{-\sigma_0 t}$ is absolutely integrable }

$$\Rightarrow \int_{-\infty}^{\infty} \left| x(t) e^{-\sigma_0 t} \right| dt < \infty$$

$$= \int_{-\infty}^{\infty} |x(t)|\, e^{-\sigma_0 t}\, dt < \infty$$

$\therefore$ If $s_0 = \sigma_0 + j\omega_0 \in ROC$

$\Rightarrow s = \sigma_0 + j\omega \in ROC \;\; \forall\, \omega$



$\therefore$ ROC is in the form of vertical strips

(poles of a transfer function)

Let $X(s)$ be a rational f^n (zeros & poles of a rational

$$\Rightarrow \quad X(s) = \frac{P(s)}{Q(s)}$$

$P(s) \rightarrow$ Roots of $P(s)$ function) : zeros

$Q(s) \rightarrow$ Roots of $Q(s)$: poles

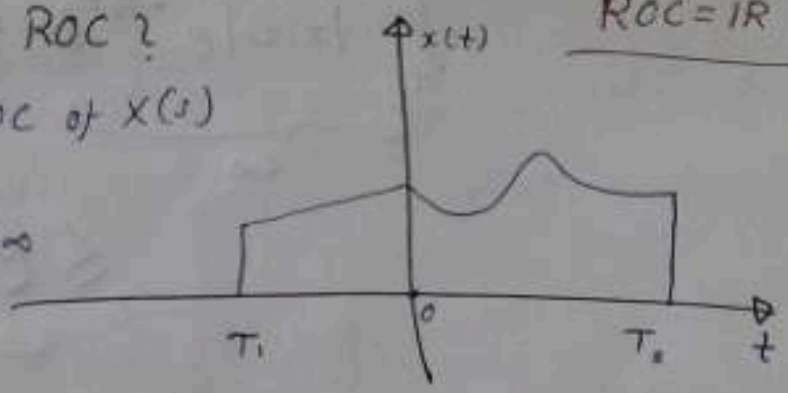
$\Rightarrow$ At poles, $X(s) \rightarrow \infty$

$\therefore$ Poles can't be part of ROC.

(II) Let $x(t)$ be finite durat^n signal

$\Rightarrow$ ROC ?

$\boxed{ROC = \mathbb{R}}$



Pf: i) Let $\sigma_0 \in$ ROC of $X(s)$

$$\Rightarrow \int_{-\infty}^{\infty} |x(t)| e^{-\sigma_0 t} dt < \infty$$

$$\Rightarrow \int_{T_1}^{T_2} |x(t)| e^{-\sigma_0 t} dt < \infty$$

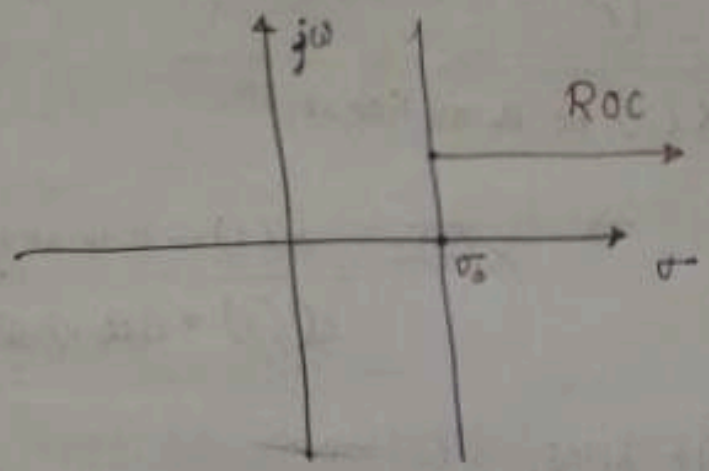
$\Rightarrow \sigma_1 = \sigma_0 + k, \quad k > 0, \quad (\text{i.e. } \sigma_1 > \sigma_0)$

$$\int_{T_1}^{T_2} \underbrace{|x(t)| e^{-\sigma_1 t} dt}_{<\infty} = \int_{T_1}^{T_2} |x(t)| e^{-(\sigma_0 + k)t} dt$$

$$= \underbrace{\int_{T_1}^{T_2} |x(t)| e^{-\sigma_0 t} e^{-kt} dt}_{<\infty} \leq \underbrace{e^{-kT_1}}_{<\infty} \underbrace{\int_{T_1}^{T_2} |x(t)| e^{-\sigma_0 t} dt}_{<\infty}$$

$(\because \sigma_0 \in ROC)$

$\therefore \sigma_1 = \sigma_0 + k \in ROC$



ii) $\|^{ly}$, for $k<0$, $\sigma_2 = \sigma_0 + k$

$$\underbrace{\int_{T_1}^{T_2} |x(t)|\, e^{-\sigma_2 t}\, dt}_{<\infty} = \int_{T_1}^{T_2} |x(t)|\, e^{-\sigma_0 t}\, e^{-kt}\, dt$$

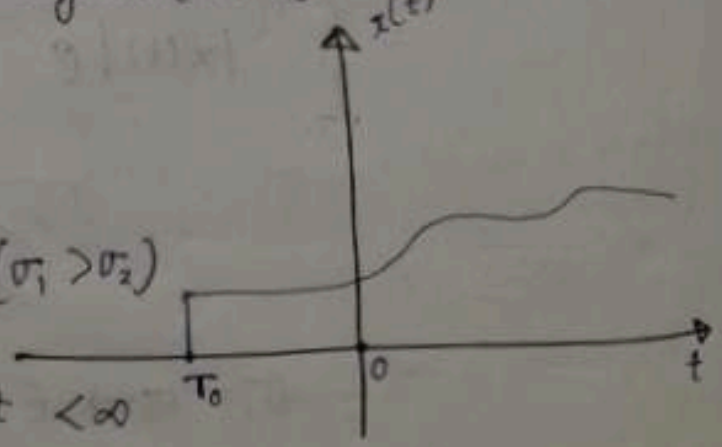
$$\leq \underbrace{e^{-kT_2}}_{<\infty} \underbrace{\int_{T_1}^{T_2} |x(t)|\, e^{-\sigma_0 t}\, dt}_{<\infty}$$

$\therefore$ $\sigma_2 = \sigma_0 + k \in ROC$, $k<\infty$

$\because$ Entire S-plane is ROC.

(III) $x(t)$ is Right Sided Signal (RSS)

ROC?



Let $\sigma_0 \in$ ROC of $X(s)$

& let $\sigma_1 = \sigma_0 + k$, $k>0$ $(\sigma_1 > \sigma_2)$

$$\Rightarrow \int_{T_0}^{\infty} |x(t)|\, e^{-\sigma_0 t}\, dt < \infty$$

$$\int_{T_0} \quad dt = \int_{T_0}^{\infty} |x(t)| e^{-(\sigma_0+K)t} dt$$

$$\leq e^{-KT_0} \int_{T_0}^{\infty} |x(t)| e^{-\sigma_0 t} dt$$

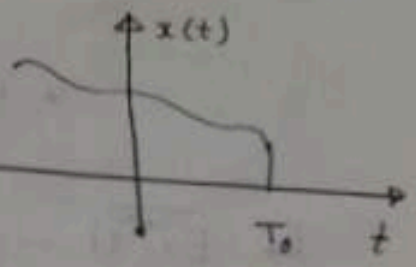
$$< \infty$$

$\therefore \sigma_1 = \sigma_0 + K \in ROC$

$\sigma_1 > \sigma_0 \in ROC$ (ROC is right sided)

(IV) For LSS $\rightarrow$ ROC, (R : $\sigma < \sigma_0$) (left sided ROC)

$$\Rightarrow \int_{-\infty}^{T_0} |x(t)| e^{-\sigma_0 t} dt < \infty$$



$\sigma_0 \in ROC$

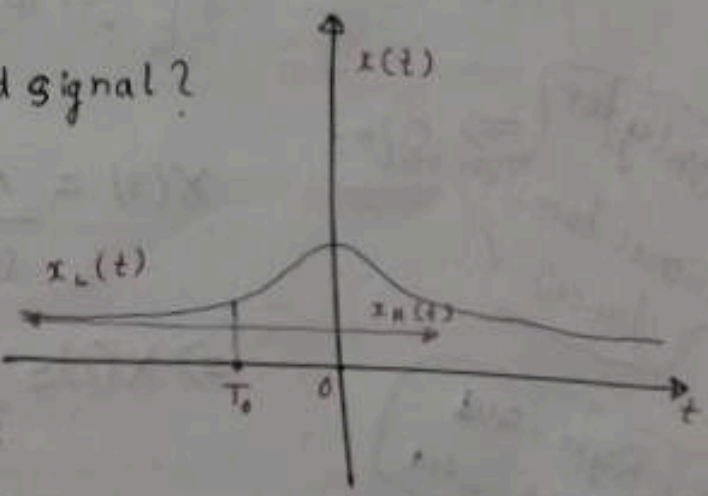
$\sigma_2 = \sigma_0 - K$

$$\Rightarrow \int_{-\infty}^{T_0} |x(t)| e^{-\sigma_0 t} e^{+Kt} dt \leq e^{+KT_0} \int_{-\infty}^{T_0} |x(t)| e^{-\sigma_0 t} dt$$

$$\leq \infty$$

(V) For bothsided signal?

ROC?

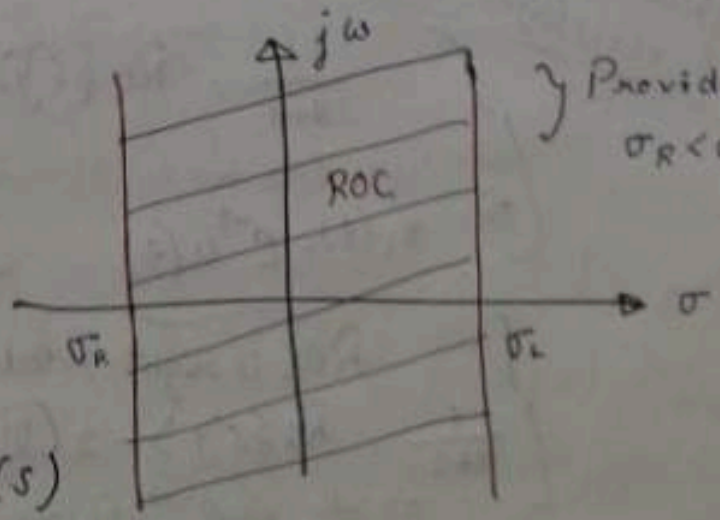


$x(t) = x_L(t) + x_R(t)$

$\downarrow$ ROC $\quad \downarrow$ ROC

$\sigma < \sigma_L \quad \sigma > \sigma_R$

Both sided = left-sided + Right-sided



} Provided $\sigma_R < \sigma_L$

$\therefore$ ROC is vertical strip (if exists)

$X(s) = X_L(s) + X_R(s)$

#

$$X(s) = \int_{-\infty}^{\infty} x(t)\, e^{-st}\, dt \quad , \text{ Inverse L.T. } , x(t) = ?$$

(Widens the class of signals for which F.T. can be found)

* Any transform which is not invertible is of no use.

* we need $X(s)$ & ROC

Ex.(1) $X(s) = \dfrac{1}{(s+1)(s+2)}$, $Re\{s\} > -1$, $x(t)$?

[Constraint: $X(s) \equiv$ rational ~~(real)~~ for any practical signal]

Only for rational transfer function $\Rightarrow$ Sol^n:

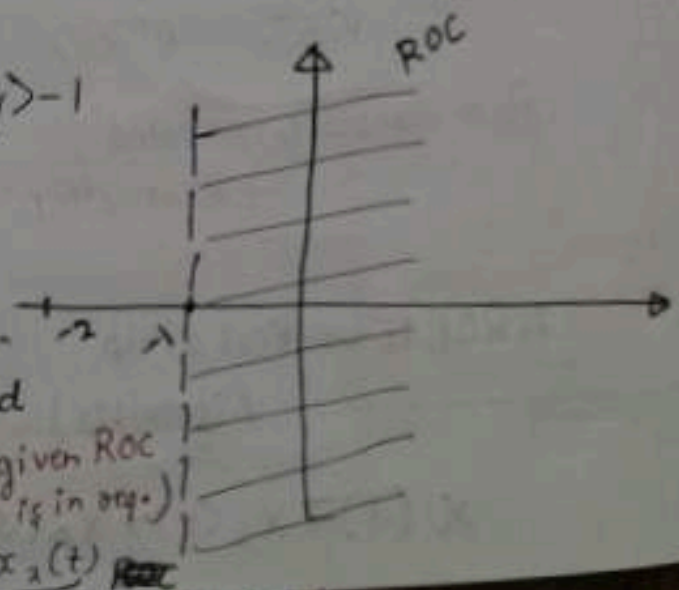
$$X(s) = \frac{A}{s+1} + \frac{B}{s+2} \Rightarrow A = 1, B = -1$$

$$\Rightarrow X(s) = \frac{1}{s+1} - \frac{1}{s+2} , \; Re\{s\} > -1$$

ROC $\equiv$ Right-sided
$x(t) \equiv$ Right-sided

$\dfrac{1}{s+1}$, $Re\{s\} > -1$

$\Rightarrow$ $\underbrace{x_1(t) = e^{-t} u(t)}_{\text{ROC is right-sided}}$



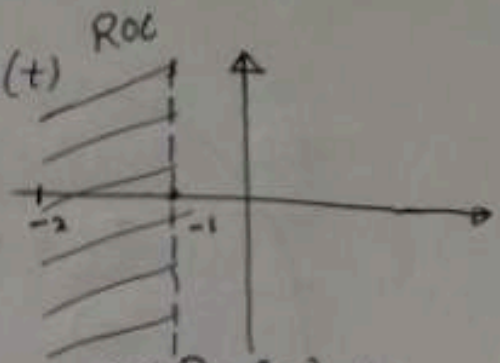
$\dfrac{1}{s+2}$, $Re\{s\} > -2$ (given ROC is in eqn.)

$\Rightarrow \dfrac{1}{s+2} \Rightarrow \underline{e^{-2t} u(t) = x_2(t)}$

$\therefore Re\{s\} > -1 \;\&\; Re\{s\} > -2$

$\therefore \; x(t) = x_1(t) - x_2(t)$

$= (e^{-t} - e^{-2t})\, u(t)$



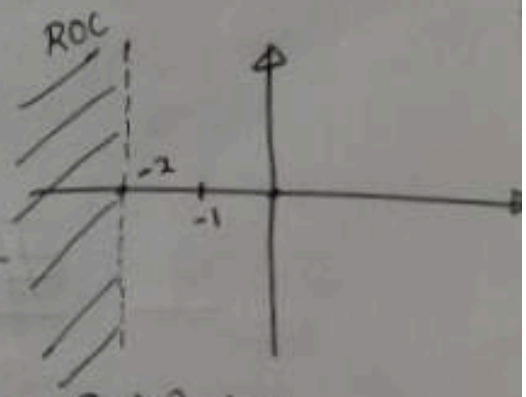
↳ ii) ROC : $-2 < \sigma < -1$

$$X(s) = \frac{1}{(s+1)(s+2)}, \quad -2 < Re\{s\} < -1$$

$$X(s) = \frac{1}{s+1} - \frac{1}{s+2}$$

$$\left\{ x(t) = -e^{-t} u(-t) - e^{-2t} u(t) \right\}$$

↳ iii) ROC : $\sigma < -2$



$$X(s) = \frac{1}{(s+1)(s+2)}$$

$$X(s) = \frac{1}{s+1} - \frac{1}{s+2}, \quad Re\{s\} < -2$$

$$\Rightarrow \; x(t) = e^{-2t} u(-t) - e^{-t} (u(-t))$$

Closed integral form

Invousion of L.T. :

$$X(s) = F.T.\left\{ x(t) e^{-\sigma t} \right\} \Rightarrow x(t) e^{-\sigma t} = I.F.T. \{X(s)\}$$

$$x(t) e^{-\sigma t} = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(s) e^{j\omega t} d\omega$$

$$\Rightarrow x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(s)\, e^{\sigma t} e^{j\omega t} d\omega$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} X(s)\, e^{(\sigma + j\omega)t}$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(s)\, e^{st}\, d\omega, \quad s = \sigma + j\omega$$

If σ is a constant.

$\Rightarrow \quad ds = j\,d\omega \Rightarrow d\omega = ds/j$

$$\Rightarrow \quad x(t) = \underbrace{\frac{1}{2\pi j} \int_{\sigma - j\omega}^{\sigma + j\omega} X(s)\, e^{st}\, ds}_{\text{Contour integral}}.$$

σ is constant $\in$ ROC

Properties of L.T.

$$ROC_{X(s)} = ROC_1 \cap ROC_2$$

can be greater than intersection

i) Linearity:

$$x_1(t) \longleftrightarrow X_1(s), R_1 \qquad x_2(t) \longleftrightarrow X_2(s), R_2$$

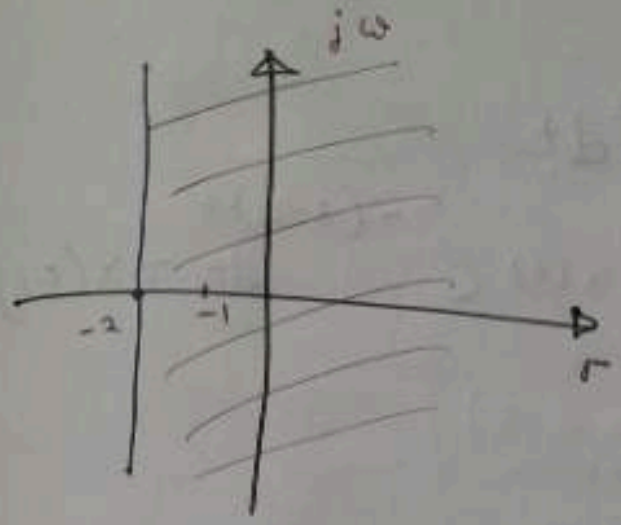
$$\alpha x_1(t) + \beta x_2(t) \longleftrightarrow \alpha X_1(s) + \beta X_2(s).$$

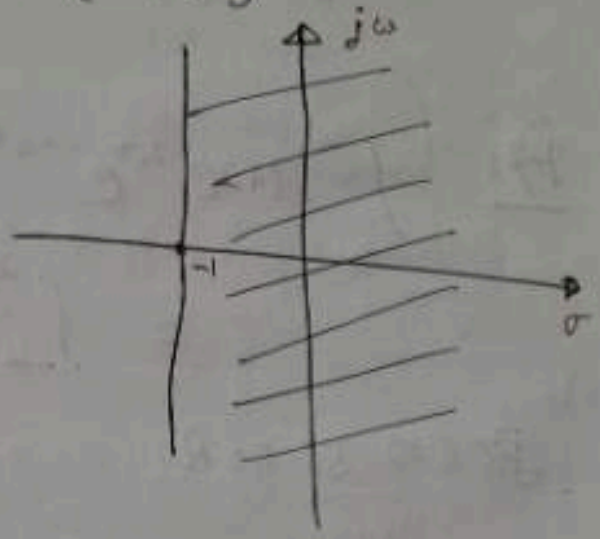
R: at least $R_1 \cap R_2$

$$X_2(s) = \frac{1}{(s+1)(s+2)}, \quad X_1(s) = \frac{1}{s+1}, \quad Re\{s\} > -1$$

$Re\{s\} > -1$

$$X_1(s) - X_2(s) = \left.\frac{+1}{(s+2)}\right\} \text{ROC}$$





ii) <u>Time Shifting</u> :

$$x(t-t_0) \longleftrightarrow X(s)\, e^{-st_0}, \quad \text{ROC: } R$$

(in a finite s-plane ROC remains same)

<u>Pf:</u> $\displaystyle\int_{-\infty}^{\infty} x(t-t_0)\, e^{-st}\, dt = \int_{-\infty}^{\infty} x(t')\, e^{-s(t'+t_0)}\, dt'$

$\Rightarrow t' = t - t_0$

$$= e^{-st_0} \underbrace{\int_{-\infty}^{\infty} x(t')\, e^{-st'}\, dt'}_{X(s)}$$

$$= X(s)\, e^{-st_0}$$

$\hookrightarrow$ For $s_0 \in R$: $\underbrace{X(s_0)}_{<\infty}\, \underbrace{e^{-s_0 t_0}}_{<\infty} \Rightarrow s_0 \in$ ROC of $X(s)e^{-st_0}$

$\therefore$ <u>ROC: R</u>

$$x(t)\, e^{s_0 t} \longleftrightarrow X(s-s_0) \quad , \quad \text{ROC}: (R+s_0) \in R$$

<u>Pf:</u> $\int_{-\infty}^{\infty} x(t) e^{s_0 t} e^{-st}\, dt$

$$= \int_{-\infty}^{\infty} x(t)\, e^{-(s-s_0)t}\, dt = X(s-s_0)$$

ROC $\Rightarrow$ $s-s_0 \in R$

iv) <u>Time Scaling:</u>

$$x(at) \longleftrightarrow \frac{1}{|a|} X\left(\frac{s}{a}\right) \quad , \quad \text{ROC}: \frac{s}{a} \in R$$

<u>Pf:</u> $\int_{-\infty}^{\infty} x(at) e^{-st} dt = \int_{-\infty}^{\infty} x(t') e^{-st'/a} \frac{dt'}{a}$, $t' = at$ $(a>0)$

$$= \frac{1}{a} X\left(\frac{s}{a}\right) , \; a>0$$

$$= \frac{-1}{a} X\left(\frac{s}{a}\right) , \; a<0$$

$$= \frac{1}{|a|} X\left(\frac{s}{a}\right) , \; \text{ROC}: \frac{s}{a} \in R$$

v) <u>Convolution:</u>

$$x_1(t) * x_2(t) \longleftrightarrow X_1(s)\, X_2(s) ,$$

ROC atleast $R_1 \cap R_2$

Pf: $y(t) \triangleq \int_{-\infty}^{\infty} x_1(\tau)\, x_2(t-\tau)\, d\tau$

$$\updownarrow$$

$$X_2(s)\, e^{-s\tau}$$

$$= \int_{-\infty}^{\infty} x_1(\tau) \frac{1}{2\pi j} \int_{\sigma-j\omega}^{\sigma+j\omega} X_2(s)\, e^{-s\tau} \cdot e^{st}\, ds$$

$$= \frac{1}{2\pi j} \int_{\sigma-j\omega}^{\sigma+j\omega} X_2(s)\, e^{st}\, ds \int_{-\infty}^{\infty} x_1(\tau)\, e^{-s\tau}\, d\tau$$

$$y(t) = \frac{1}{2\pi j} \int_{\sigma-j\omega}^{\sigma+j\omega} \underbrace{\boxed{X_1(s)\, X_2(s)}}_{Y(s)}\, e^{st}\, ds$$

$$\Rightarrow y(t) = x_1(t) * x_2(t) \longleftrightarrow Y(s) = X_1(s) \cdot X_2(s)$$

ROC $\Rightarrow$ Ex: Let $X_1(s) = \dfrac{P(s)}{Q(s)}$, $X_2(s) = \dfrac{Q(s)}{P(s)}$

$X_1(s) \cdot X_2(s) = 1 \;\forall s$ $\therefore$ ROC is entire

S-plane $\supset R_1 \cap R_2$

vi) Differentiation:

$$\frac{dx(t)}{dt} \longleftrightarrow s\,X(s) \quad , \text{ROC: at least } R$$

$$x(t) = \frac{1}{2\pi j} \int_{\sigma-j\omega}^{\sigma+j\omega} X(s)\, e^{st}\, ds \Rightarrow \frac{dx(t)}{dt} = \frac{1}{2\pi j} \int_{\sigma-j\omega}^{\sigma+j\omega} sX(s)\, e^{st}\, ds$$

$$\Rightarrow \frac{dx(t)}{dt} \longleftrightarrow s\,X(s)$$

vii) $t\,x(t) \longleftrightarrow -\dfrac{dX(s)}{ds}$, ROC : R ($\because$ diff. of $X(s)$ doesn't introduce a pole at new location)

Pf: $X(s) = \int_{-\infty}^{\infty} x(t)\, e^{-st}\, dt$

$$\frac{dX(s)}{ds} = \int_{-\infty}^{\infty} x(t)(-t)\, e^{-st}\, dt \Rightarrow -t\,x(t) \longleftrightarrow \frac{dX(s)}{ds}$$

i.e. $t\,x(t) \longleftrightarrow \dfrac{-dX(s)}{ds}$

$$\boxed{\begin{array}{l} \Rightarrow \dfrac{P(s)}{Q(s)} \Rightarrow d(P) >= d(Q) \\ \qquad\qquad \text{prop : (vi)} \\ \Rightarrow \text{repeated poles} \Rightarrow \\ \qquad\qquad \text{prop : (vii)} \end{array}}$$

Ex: $\delta(t) \longleftrightarrow 1$

$\Rightarrow \dfrac{d}{dt}\,\delta(t) \longleftrightarrow s$

$\Rightarrow \dfrac{d^2}{dt^2}\,\delta(t) \longleftrightarrow s^2$

$\vdots$

Useful in inversion of improper rational $X(s)$

Ex: $e^{-at}u(t) \longleftrightarrow \dfrac{1}{s+a}$, $\sigma > -a$

$\Rightarrow t e^{-at}u(t) \longleftrightarrow -\dfrac{(-1)}{(s+a)^2} = \dfrac{1}{(s+a)^2}$, $\sigma > -a$

$\dfrac{t^2}{2} e^{-at} u(t) \longleftrightarrow \dfrac{1}{(s+a)^3}$, $\sigma > -a$

$\vdots$

$$\frac{t^{n-1}}{(n-1)!} e^{-at} u(t) \longleftrightarrow \frac{1}{(s+a)^n}, \quad \sigma > -a$$

$\int_{-\infty}^{t} x(\tau)\, d\tau \longleftrightarrow \frac{X(s)}{s}$, ROC: atleast $R \cap \{Re\{s\} > 0\}$

$$= x(t) * u(t) \longleftrightarrow X(s)U(s)$$

$$= X(s) \times \frac{1}{s}$$

$$\left\{ u(t) = e^{-at} u(t) \longleftrightarrow \frac{1}{s+0}, \; Re\{s\} > 0 \right\}$$

$$\therefore U(s) = \frac{1}{s}, \quad Re\{s\} > 0$$

Useful L.T. Pairs :

i) $\cos(\omega_0 t)\, u(t) \longleftrightarrow \frac{s}{s^2+\omega_0^2}$, $Re\{s\} > 0$

$\left(\text{Hint} : \cos(\omega_0 t) = \frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2} \right)$

ii) $\sin(\omega_0 t)\, u(t) \longleftrightarrow \frac{\omega_0}{s^2+\omega_0^2}$, $Re\{s\} > 0$

iii) $e^{-at} \cos(\omega_0 t)\, u(t) \longleftrightarrow \frac{s+a}{(s+a)^2+\omega_0^2}$, $Re\{s\} > -a$

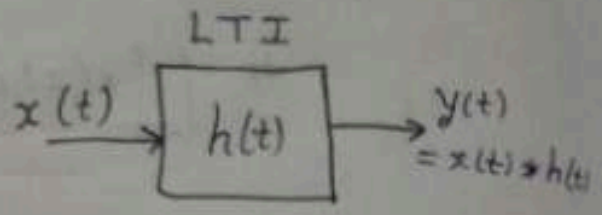
iv) $e^{-at} \sin(\omega_0 t)\, u(t) \longleftrightarrow \frac{\omega_0}{(s+a)^2+\omega_0^2}$, $Re\{s\} > -a$

v) $e^{-a|t|} \longleftrightarrow \frac{2a}{a^2-s^2}$, $-a < Re\{s\} < a$, provided $a > 0$

Hint: $e^{-a|t|} = e^{-at}$

LTI System Analysis Using L.T.

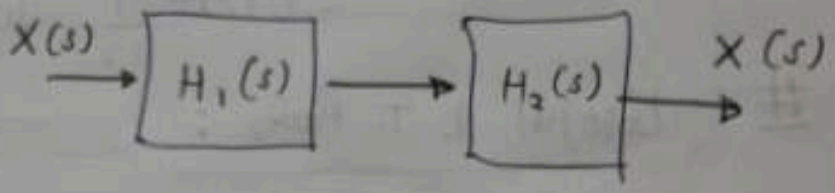
1) response
2) inverse
3) identifying system



$$Y(s) = X(s)\,H(s)$$

Used for: i) Finding o/p $y(t)$

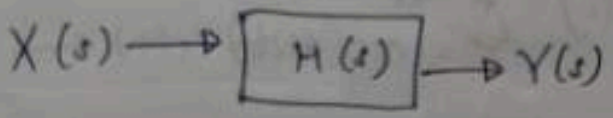
ii) Finding inverse system:



$$\Rightarrow H_1(s)\,H_2(s) = 1 \Rightarrow$$

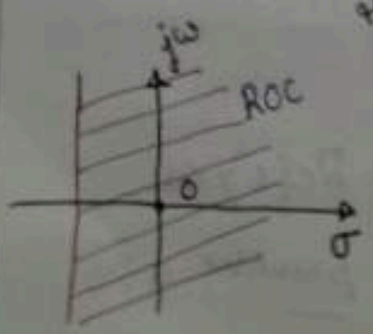
$$H_2(s) = \frac{1}{H_1(s)}$$

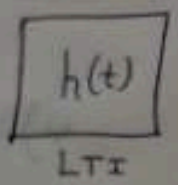
iii) System Identification:



$$\Rightarrow H(s) = Y(s)/X(s)$$

$\Rightarrow$ Stable & Causal System:





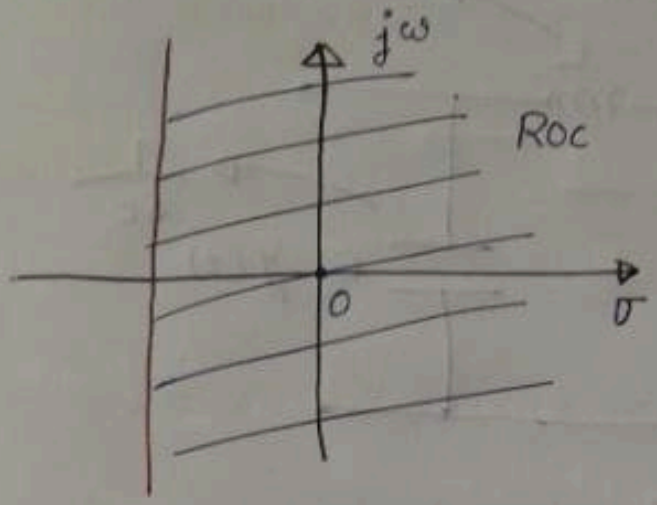
i) Let LTI system be stable

$$\Leftrightarrow \int_{-\infty}^{\infty} |h(t)|\,dt < \infty$$

$$\Leftrightarrow \int_{-\infty}^{\infty} |h(t)|\,e^{-\sigma t}\,dt \Leftrightarrow \sigma = 0 \in \text{ROC of } H(s)$$

$$\Leftrightarrow j\omega \in \text{ROC of } H(s)$$

ii) Let LTI be causal $\Rightarrow h(t) = 0 \; \forall \; t<0$



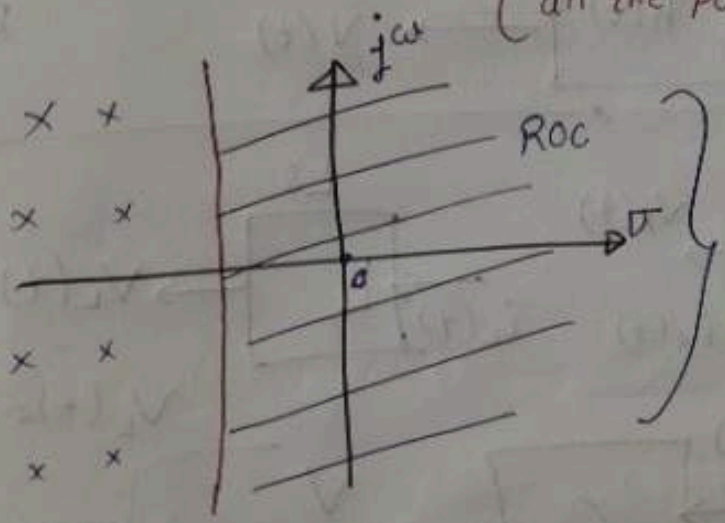
$\Rightarrow h(t)$ is RSS.

$\Rightarrow$ ROC of $H(s)$ is right sided.

(towards right of rightmost pole)

$\Rightarrow$ Now, if LTI system is both stable & causal.

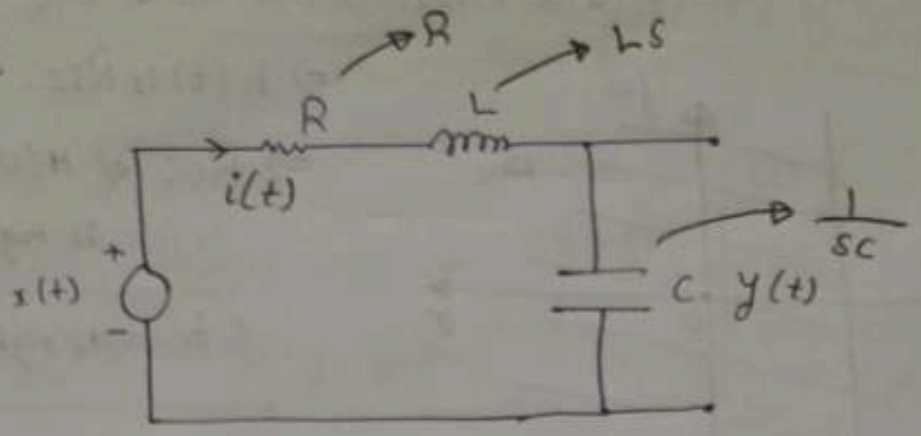
(all the poles must be on the Left half s-plane)



$\Rightarrow$ All poles must be in left half of s-plane.

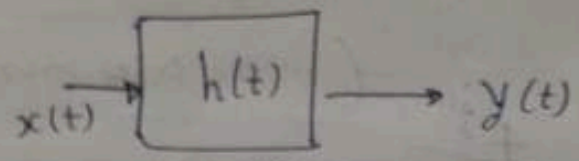
(For stable and causal LTI)

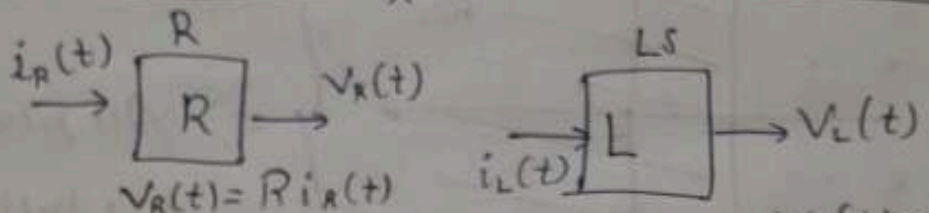
Q.9.



H(s)

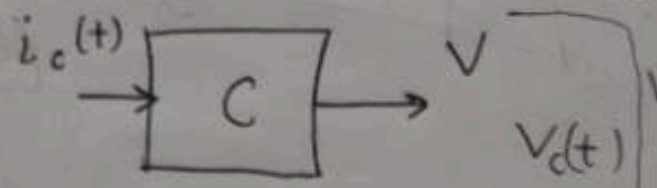
$$H(s) = \frac{1/CS}{R + LS + 1/CS}$$





$V_R(t) = R\, i_R(t)$

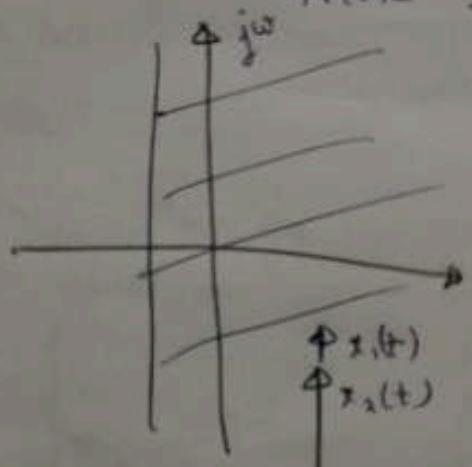
$V_L(t) = L \dfrac{di_L(t)}{dt}$



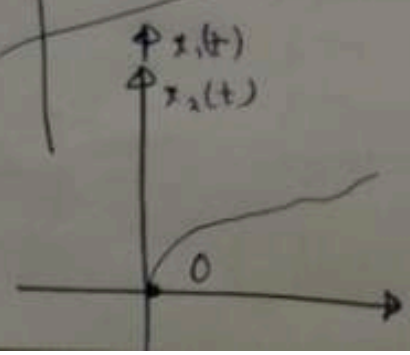
$V_L(s) = LS\, I_L(s)$

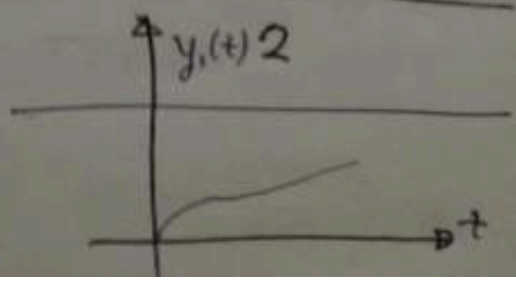
$\dfrac{V_L(s)}{I_L(s)} = LS$

$$H(s) = \frac{1}{RCS + LCS^2 + 1} = \frac{1/LC}{\left(S^2 + \frac{RS}{L} + \frac{1}{LC}\right)}$$



$$S = -\frac{R}{L} \pm \sqrt{\frac{R^2}{L^2} - \frac{4}{LC}}$$





(6)

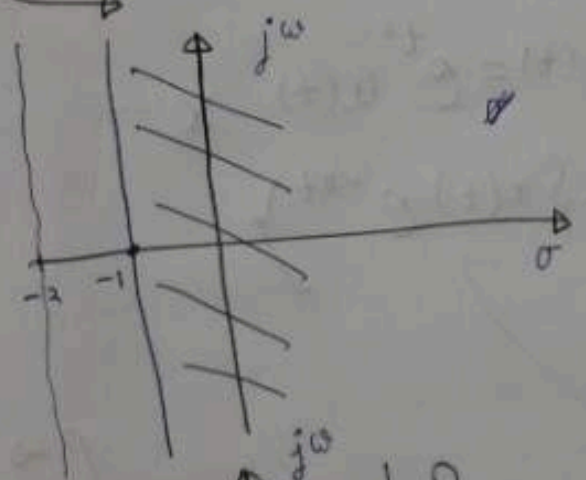
$$X(s) = \frac{s+2}{s-2}, \quad x(t)=0, \quad t>0$$

$$y(t) = \frac{-2}{3} e^{2t} u(-t) + \frac{1}{3} e^{-t} u(t)$$

$$Y(s) = \underbrace{\frac{2}{3} \frac{1}{s-2}}_{\sigma<2} + \underbrace{\frac{1}{3} \frac{1}{s+1}}_{\sigma>-1}$$

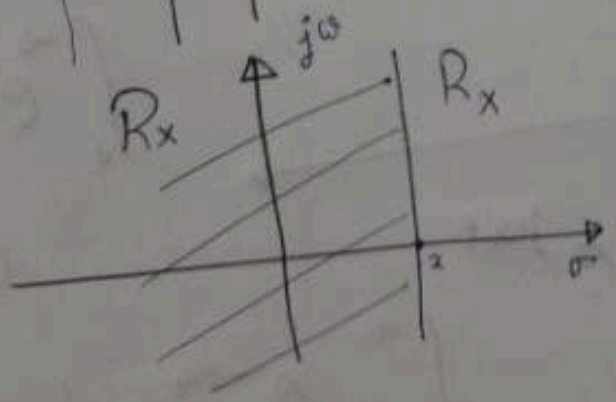
$$= \frac{s}{(s-2)(s+1)}$$

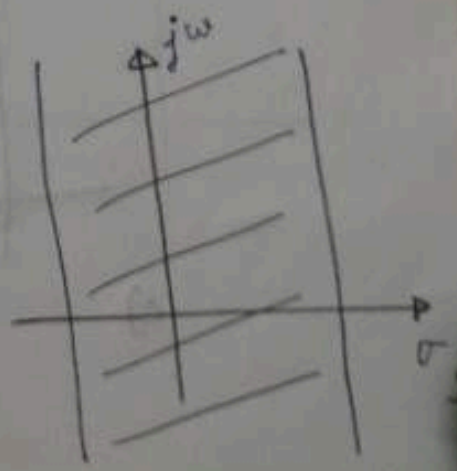
$$H(s) = \frac{Y(s)}{X(s)} = \frac{s}{(s+1)(s+2)}$$



$$Y(s) = H(s)X(s)$$

$$R_Y = R_H \cap R_x$$





$R_H > -1$

c)

$$x(t)e^{st} \longrightarrow \boxed{h(t)} \longrightarrow Y(t) = \int h(\tau) e^{s(t-\tau)} d\tau$$

LTI

$$= e^{st} \int h(\tau) e^{-s\tau} d\tau$$

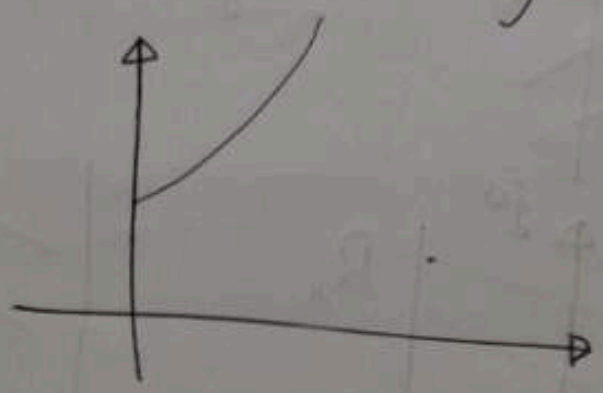
$$e^{3t} \longrightarrow H(3)\, e^{3t}$$

$$\frac{3}{20} e^{3t}$$

4) b)

$$x(t) = e^{t^2} u(t) \quad \times$$

$$F.T.\{x(t) e^{-\sigma t}\}$$



$$\int_0^\infty e^{t^2 - \sigma t} dt \not< \infty$$

$$e^{t(t-\sigma)}$$

c) $e^{j\omega_0 t}$

$$\int_{-\infty}^{\infty} e^{-\sigma t} dt \not< \infty$$